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Lighting Device

Abstract

A lighting device for use by a person comprising an inner bracket and an outer bracket. The device further comprises a first light rotatably engaged the inner bracket and the outer bracket. The device further comprises a second light rotatably engaged with the inner bracket and the outer bracket. The second light is disposed below the first light. The device further comprises a third light engaged with the inner bracket and the outer bracket. The third light is disposed below the second light.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to and the benefit of U.S. Provisional Application Ser. No. 63/559,761 filed on Feb. 29, 2024; U.S. Provisional Application Ser. No. 63/676,418 filed on Jul. 28, 2024; U.S. Provisional Application Ser. No. 63/678,858 filed on Aug. 2, 2024; U.S. Provisional Application Ser. No. 63/682,314 filed on Aug. 12, 2024; and U.S. Provisional Application Ser. No. 63/686,607 filed on Aug. 23, 2024.

BACKGROUND OF THE INVENTION

[0002] Light therapy employing light emitting diodes have recently been used to trat facial conditions such as acne and other conditions. There is a continuing need for ergonomic devices to deliver such light therapy.

SUMMARY OF THE INVENTION

[0003] A light device for use by a person comprising an inner bracket and an outer bracket. The device further comprises a first light rotatably engaged the inner bracket and the outer bracket. The device further comprises a second light rotatably engaged with the inner bracket and the outer bracket. The second light is disposed below the first light. The device further comprises a third light engaged with the inner bracket and the outer bracket. The third light is disposed below the second light. The first light, second light, and third light each comprise a plurality of light sources that in the preferred embodiment are light emitting diodes having different types of colors to allow for different types of light treatment therapy.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The following description of the invention will be more fully understood with reference to the accompanying drawings in which:

[0005] FIG. **1** is a perspective view of a lighting device according to the present invention mounted to a stand in a first position;

[0006] FIG. **2** is a perspective view of the lighting device mounted to the stand in a second position;

[0007] FIG. **3** is a top perspective view of a light unit of the lighting device in a position selected by a person shown comprising a first light and a second light rotatably engaged with an inner bracket and an outer bracket of the light unit, and a third light engaged with the inner bracket and outer bracket of the light unit;

[0008] FIG. **4** is a bottom perspective view of the first light and second light rotatably engaged with the inner bracket and outer bracket of the light unit and the third light engaged with the inner bracket and outer bracket of the light unit;

[0009] FIG. **5** is a cross-section view of the first light and second light rotatably engaged with the inner bracket of the light unit and the third light engaged with the inner bracket of the light unit;

[0010] FIG. **6** is a cross-section view of the first light and the second light rotatably engaged with the outer bracket of the light unit and the third light engaged with the outer bracket of the light unit; and

[0011] FIG. **7** is a high level block schematic of an electronic circuit for controlling the light emitting diodes of the first light, the second light, and the third light.

DESCRIPTION OF THE INVENTION

[0012] Referring to FIGS. **1-2**, a lighting device **10** according to the present invention is shown for use by a person or practitioner such as a cosmetologist to provide a plurality of LED lights for a variety of purposes and applications during cosmetic procedures such as LED light therapy with different colors for treating various facial and/or other skin ailments. Lighting device **10** generally comprises a light unit **100** rotatably engaged with a stand **14**.

[0013] With continued reference to FIGS. **1** and **2**, stand **14** comprises a base **16** and a plurality of wheels **18** mounted to the bottom of base **16** to allow lighting device **10** to be easily moved. Stand **14** further comprises a tubular pole **20** attached to base **16** by conventional means and a tubular pole **22** telescopically disposed and moveable with tubular pole **20** by conventional means. Stand **14** further comprises a bracket **24** comprising a lower end **26** engaged with tubular pole **22** and an upper end **28**. Stand **14** further comprises a tubular pole **30** comprising a lower end **32** rotatably engaged with upper end **28** of bracket **24**. Tubular pole **30** further comprises a collar **34** for locking a tubular pole **36** in a position selected by the person. Tubular pole **36** is telescopically engaged within tubular pole **30** and can be locked in place by tightening collar **34**. Tubular pole **36** further comprises an upper end **38** that is securely engaged with a mounting bracket **40**. Mounting bracket **40** comprises a lower end **42** engaged with upper end **38** of tubular pole **36** and an upper end **44** that is rotatably engaged with light unit **100**. Stand **14** and all of its components are well known in the art and widely available.

[0014] With continued reference to FIGS. **1** and **2**, light unit **100** comprises an inner bracket **102** and an outer bracket **140**. Light unit **100** further comprises a first light **180** rotatably engaged with inner bracket **102** and outer bracket **140**. Light unit **100** further comprises a second light **240** rotatably engaged with inner bracket **102** and outer bracket **140**. Light unit **100** further comprises a third light **280** engaged with inner bracket **102** and outer bracket **140**. Second light **240** is disposed below first light **180**, and third light **280** is disposed below second light **240**. First light **180** and second light **240** can be rotated outward from third light **280** to form a partial or full dome shape as shown in FIG. **2**. First light **180**, second light **240**, and third light **280** can be placed in many positions such as a vertical overlapping position as shown in FIG. **1**. First light **180** and second light **240** can be rotated to many different positions other than as shown in FIGS. **1** and **2**.

[0015] Referring to FIGS. **3** and **4**, first light **180** comprises a housing **182** comprising a rear wall comprising a median portion **186**, a left end **188** rotatably engaged with inner bracket **102**, and a right end **192** rotatably engaged with outer bracket **140**. Left end **188** is curved downward and inward from median portion **186**. Right end **192** is curved downward and inward from median portion **186** towards left end **188**. Housing **182** further comprises a circular shaped opening **190** (not shown) extending completely thru left end **188** to allow left end **188** of housing **182** to rotatably engage with inner bracket **102**. Housing **182** further comprises a circular shaped opening **194** (not shown) extending completely thru right end **192** to allow right end **192** of housing **182** to rotatably engage with outer bracket **140**. First light **180** further comprises a plurality of light sources **210** disposed within housing **182** substantially extending from left end **188** to right end **192**. In the embodiment shown, light sources **210** are light emitting diodes. First light **180** further comprises a printed circuit board **214** disposed within housing **182** substantially extending from left end **188** to right end **192** for mounting and electrically connecting light sources **210** by conventional means. First light **180** further comprises a transparent cover **220** attached to housing **182** by conventional means such as grooves disposed in the side walls of housing **182**. Light sources **210** are disposed below transparent cover **220** such that light from light sources **210** pass thru transparent cover **220**. Transparent cover **220** can take other forms such as frosted.

[0016] With continued reference to FIGS. **3** and **4**, second light **240** comprises a housing **242** comprising a rear wall **244** comprising a median portion **246**, a left end **248** rotatably engaged with inner bracket **102** and a right end **252** rotatably engaged with outer bracket **140**. Left end portion **248** is curved downward and inward from median portion **246**. Right end portion **252** is curved downward and inward from median portion **246** towards left end portion **248**. Housing **182** further comprises a circular shaped opening **250** (not shown) extending completely thru left end **248** to allow left end **248** of housing **242** to rotatably engage with inner bracket **102**. Housing **182** further comprises a circular shaped opening **254** (not shown) extending completely thru right end **252** to allow right end **252** of housing **182** to rotatably engage with outer bracket **140**. Second light **240** further comprises a plurality of light sources **264** disposed within housing **242** substantially extending from left end **248** to right end **252**. In the embodiment shown, light sources **264** are light emitting diodes. Second curved light **240** further comprises a printed circuit board **268** disposed within housing **242** substantially extending from left end **248** to right end **252** for mounting and electrically connecting light sources **264** by conventional means. Second light **240** further comprises a transparent cover **272** attached to housing **242** by conventional means such as grooves disposed in the side walls of housing **242**. Light sources **264** are disposed below transparent cover **272** such that light from light sources **264** passes thru transparent cover **272**. Transparent cover **272** may take other forms such as frosted.

[0017] With continued reference to FIGS. **3** and **4**, third light **280** comprises a housing **282** comprising a rear wall **284** comprising median portion **286**, a left end **288** rotatably engaged with inner bracket **102** and a right end **292** rotatably engaged with outer bracket **140**. Left end portion **288** of rear wall **284** is secured or fixed to inner bracket **102** by a plurality of screws (not shown). In the embodiment shown, third light **280** does not rotate relative to inner bracket **102**. In other embodiments, third curved light **280** may rotate relative to inner bracket **102**. Left end **288** is curved downward and inward from median portion **286**. Right end **292** is curved downward and inward from median portion **286** towards left end portion **288**. Third light **280** further comprises a plurality of light sources **300** disposed within housing **282** substantially extending from left end **288** to right end **292**. In the embodiment shown, light sources **300** are light emitting diodes. Third light **280** further comprises a printed circuit board **310** disposed within housing **282** substantially extending from left end **288** to right end **292** for mounting and electrically connecting light sources **300** by conventional means. Third light **280** further comprises a transparent cover **320** attached to housing **242** by conventional means such as grooves disposed in the side walls of housing **282**. Light sources **300** are disposed below transparent cover **320** such that light from light sources **300** passes thru transparent cover **272**. Transparent cover **3202** may take other forms such as frosted. As shown in FIG. **3**, as will be described more fully herein, third light **280** further comprises a series of buttons **322**, **324**, and **326** disposed on median portion **286** of rear wall **284** of housing **282** that when depressed control the on/off state, color, and/or brightness of light sources **210** of first light **180**, light sources **264** of second light **240**, and/or light sources **300** of third light **280**.

[0018] Referring to FIG. **5**, a cross-section view shows first light **180** and second light **240** rotatably engaged with inner bracket **102** and third light **280** engaged with inner bracket **102**. Inner bracket **102** comprises a left part **104** adapted to rotatably engage with bracket **40** of stand **12**. Left part **104** comprises a channel **106** for wiring and an opening **108** for rotatably mounting with bracket **40**. Inner bracket **102** further comprises a right part **110** adapted to rotatably receive first light **180** and second light **240** and engage with third light **280**. Right part **110** comprises an outermost annular step or surface **112**, a middle annular step or surface **114**, and an innermost annular step or surface **116**. Opening **190** of first light **180** is disposed about outermost annular surface **116** and middle annular surface **114** of right part **110**. A spacer **128** is provided adjacent first light **180**. Opening **250** of second light **240** is disposed about middle annular surface **114** and innermost annular surface **116** of right part **110**. Right part **140** further comprises a hole **124** so that third light **280** can be screwed to right part **110** by a screw (not shown). Although one (1) hole is shown, three (3) holes and screws are employed so that third light **280** is fixed to right part **152**. A spacer **130** is provided adjacent second light **240** and third light **280**. An oaring **120** is disposed between outermost annular surface **112** of right part **110** and left end **196** of housing **182** of first light **180** to reduce friction during rotation. An O-rings **122** is disposed between middle annular surface **114** of right part **110** and left end **246** of housing **242** of second light **240** to reduce friction during rotation. A retractable pin **118** is provided in left end **196** of housing **182** of first curved light **180** that removably engages and/or locks with an indentation **132** of right part **110** of inner bracket **102** to secure first light **180** in a selected position. Multiple indentations may be provided so first light **180** can be rotated to different positions. A retractable pin **119** is provided in left end **256** of housing **242** of second light **240** that removably engages and/or locks with an indentation **134** of inner bracket **102** to secure second light **240** is a selected position. Multiple indentations may be provided so second light **180** can be rotated to different positions. Pin **118** and pin **119** are well known in the art and widely available.

[0019] Referring to FIG. **6**, a cross-section view shows first light **180** and second light **240** rotatably engaged with outer bracket **140** and third light **280** engaged with outer bracket **140**. Outer bracket **140** comprises a cap or flange **142** and a left part **152** adapted to rotatably engage with first light **180** and second light **240** and engage with third light **280**. Left part **152** comprises a channel **170** for passing of wiring from third light **280** to first light **180** and second light **240**. Wires **332** are provided to electrically connect an electronic circuit **400** (FIG. **4**—to be described) of third light **280** with light sources **210** of first light **180**. Wires **330** are provided to electrically connect electronic circuit **400** of third light **280** with light sources **264** of second light **240**. Left part **152** comprises an outermost annular step or surface **154**, a middle annular step or surface **156**, and an innermost annular step or surface **158**. Opening **194** of first light **180** is inserted and disposed about outmost annular surface **154** and middle annular surface **156** of left part **152**. A spacer **160** is provided adjacent first light **180**. Opening **254** of second light **240** is inserted and disposed about middle annular surface **156** and innermost annular surface **158** of left part **152**. Left part **152** further comprises a hole **168** so that third light **280** can be screwed to left part **152** by a screw (not shown). Although one (1) hole is shown, three (3) holes and screws are employed so that third light **280** is fixed to left part **152**. A spacer **162** is provided adjacent second light **240** and third light **280**. An o-ring **166** is disposed between innermost annular surface **154** of left part **152** and right end **198** of housing **182** of first light **180** to reduce friction during rotation. An o-ring **164** is disposed between middle annular surface **156** of left part **152** and right end **258** of housing **242** of second light **240** to reduce friction during rotation. A pin **172** is provided in right end **198** of housing **182** of first curved light **180** that removably engages and/or locks with an indentation **176** of left part **152** of outer bracket **140** to secure first light **180** is a selected position. Multiple indentations may be provided so first light **180** can be rotated to different positions. A pin **174** is provided in right end **258** of housing **242** of second light **240** that removably engages and/or locks with an indentation **178** of outer bracket **140** to secure second light **240** is a selected orientation. Multiple indentations may be provided so second light **240** can be rotated to different positions. Pins **172** and **174** are well known in the art and widely available.

[0020] Third light **280** further comprises an electronic circuit **400** (FIG. **4**) connected to printed circuit board **310**, or another printed circuit board, to power and control operation of light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280** such as light color and/or temperature. Light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280** may have the same or different color and/or temperatures depending upon the desired application and/or treatment. Electronic circuit **400** of third light **280** is electrically connected to printed circuit board **214** of first light **180** and printed circuit board **268** of second light **240** by conventional wiring as previously described. Electronic circuits comprising microcontrollers to control the color and/or temperature of a single set or multiple arrays of Light emitting diodes are well known in the art. By way of example only, an electronic circuit for controlling light emitting diodes is described in U.S. Pat. No. 9,648,217 that is hereby incorporated into this specification in its entirety.

[0021] Referring to FIG. **7**, where a preferred embodiment of electronic circuit **400** is generally shown. Electronic circuit **400** comprises a microcontroller **402** that may be programmed by well known means in the art to activate and/or control light emitting diodes **210** of first curved light **180**, light emitting diodes **264** of second curved light **240** and light emitting diodes **300** of third curved light **280**. Microcontroller **402** is connected to a LED driver **404** that is connected to a red LED array **414**. Microcontroller **402** is connected to a LED driver **406** that is connected to a blue LED array **416**. Microcontroller **402** is connected to a LED driver **408** that is connected to an infrared LED array **418**. Microcontroller **402** is connected to a LED driver **410** that is connected to a white LED array **420**. Microcontroller **402** is connected to a LED driver **412** that is connected to a warm white LED array **420**. Each of light sources **210** of first light **180**, light sources **264** of second light **240** and light sources **300** of third light **280** are made up of red LED array **414**, blue LED array **416**, infrared LED array **418**, white LED array **420**, and warm white LED array **420** and electrically connected in the same manner. Electronic circuit **400** further comprises a first switch **424** connected to microcontroller **402**. First switch **424** is activated by depression of button **322** that sends a signal to microcontroller **402** to turn on and off the light sources of first light **180**, second light **240**, and third light **280**. Electronic circuit **400** further comprises a second switch **426** connected to microcontroller **402**. Second switch **426** is activated by depression of button **324** that sends a signal to microcontroller **402** to change the color mode of the light sources of first light **180**, second light **240**, and third light **280** in any programmed color. Electronic circuit **400** further comprises a third switch **428** connected to microcontroller **402**. Third switch **428** is activated by depression of button **326** that sends a signal to microcontroller **402** to adjust the brightness of the light sources of first light **180**, second light **240**, and third light **280** in any predetermined level. Electronic circuit **400** further comprises a USB Type C connector **430** connected to microcontroller **402** by a voltage regulator **432** that drops the 36V input to 3.3V for microcontroller **402**. Device **10** further comprises a conventional electrical cord (not shown) having a conventional AC/DC power supply (not shown) that a person can plug into a common electrical wall outlet and an USB connector that connects with USB connector **430** disposed on the bottom pole of stand **14** or many other places on stand **14** or light unit **100**. Conventional wiring is provided thru the poles and brackets to connect USB connector **430** with and provide power to printed circuit board **310** of third curved light **280**. Such electrical cords having a power supply are well known in the art and widely available as are USB connectors.

[0022] With reference to FIG. **3** and FIG. **7**, button **326** is used to turn on and off lighting device **10**. Depression and hold for two (2) seconds of button **326** causes a signal from switch **424** to be sent to microcontroller **402** of electronic control circuit **400** to turn on or off light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280**.

[0023] With continued reference to FIG. **3** and FIG. **7**, button **322** is used to select the color mode of light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280**. The default mode at startup (power turned on) are the colors red and infrared by activation of light emitting diodes **414** and **418** of the light sources **210**, **264**, and **300**. Pressing button **322** causes a signal from switch **426** to be sent to a microcontroller **402** of electronic control circuit **400** of third light **280** to activate the next color mode, namely, blue light emitting diodes **416** of light sources diodes **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280**. Pressing button **322** again causes microcontroller **402** of electronic control circuit **400** of third light **280** to activate the next color mode, namely, “warm white light” light emitting diodes of **422** of light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third curved light **280**. Pressing button **322** again causes microcontroller **402** of electronic control circuit **400** of third light **280** to activate the next color mode, namely, “cold white light” light emitting diodes of **420** of light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280**. Cycle thereafter.

[0024] With continued reference to FIG. **3** and FIG. **7**, pressing of button **324** adjusts the brightness of light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280**. Different brightness are used from low to high cycle.

[0025] In other embodiments, any presently or futurely developed light source having therapeutic properties may be used instead of a light emitting diode light source. By way of further example only, stand **14** may be replaced with any other type of mounting means, such as table mounted means, with relocation of USB connector **430**.
[0026] The foregoing description is intended primarily for purposes of illustration. This invention may be embodied in other forms or carried out in other ways without departing from the spirit or scope of the invention. Modifications and variations still falling within the spirit or scope of the invention will be readily apparent to those of skill in the art.

Claims

1. A device for use by a person comprising: an inner bracket; an outer bracket; a first light rotatably engaged with said inner bracket and said outer bracket; a second light rotatably engaged with said inner bracket and said outer bracket; said second light being disposed below said first light; and a third light engaged with said inner bracket and said outer bracket; said third light being disposed below said second light.
2. The device of claim 1, wherein said first light comprises a housing; said housing comprises a rear wall comprising a median portion, a first end, a second end, a first opening and a second opening thru said first end and said second end, respectively; said opening of said first end of said first light being rotatably engaged with said inner bracket; said opening of said second end of said first light being rotatably engaged with said outer bracket.
3. The device of claim 2, wherein said second light comprises a housing; said housing comprises a rear wall comprising a median portion, a first end, a second end, and a first opening and a second opening thru said first end and said second end, respectively; said opening of said first end of said second light being rotatably engaged with said inner bracket; said opening of said second end of said second light being rotatably engaged with said outer bracket.
4. The device of claim 3, wherein said third light comprises a housing; said housing comprises a rear wall comprising a median portion, a first end, and a second end; said first end of said third light is securely engaged with said inner bracket; and second end of said third light is securely engaged with said outer bracket.
5. The device of claim 4, further comprising a first annular spacer disposed about said inner bracket and disposed between said first end of said first light and said first end of said second light.
6. The device of claim 5, further comprising a second annular spacer disposed about said inner bracket and disposed between said first end of said second light and said first end of said third light.
7. The device of claim 6, wherein said inner bracket comprises an outermost annular step engaged with said first light.
8. The device of claim 7, wherein said inner bracket further comprises a middle annular step engaged with said second light.
9. The device of claim 1, wherein said left end of said rear wall of said first light is curved downward and inward from said median portion of said rear wall of said first light; said right end of said rear wall of said first light is curved downward and inward from said median portion of said rear wall of said first light towards said left end portion of said rear wall of said first light.
10. The device of claim 9, wherein said left end of said rear wall of said second light is curved downward and inward from said median portion of said rear wall of said first light; said right end of said rear wall of said first light is curved inward from said median portion of said rear wall of said first light towards said left end of said rear wall of said first light.
11. The device of claim 1, wherein said first light comprises a plurality of light sources.
12. The device of claim 11, wherein each of said light sources of said first light is a light emitting diode.

13. The device of claim 12, wherein said second light comprises a plurality of light sources.
14. The device of claim 13, wherein each of said light sources of said second light is a light emitting diode.
15. The device of claim 13, wherein said third light comprises a plurality of light sources.
16. The device of claim 1, wherein each of said light sources of said third light is a light emitting diode.
17. The device of claim 1, wherein said inner bracket is rotatably connected to a stand.